Questions

Q1.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Figure 7 shows bubbles of air that a diver breathes out.

The bubbles rise towards the surface.

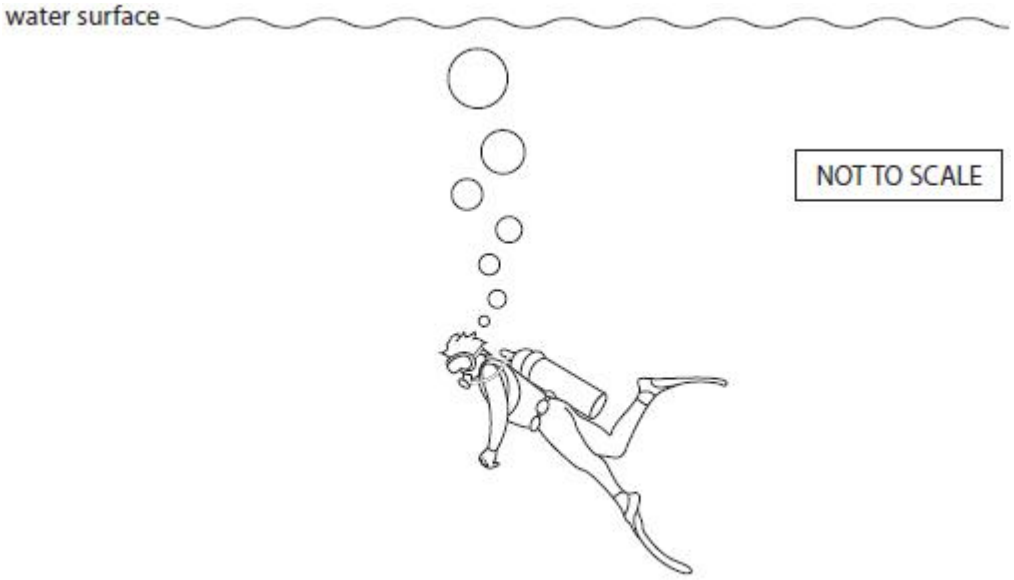


Figure 7

(i) Which row of the table is correct as one bubble rises?

(1)

		the air pressure in the bubble	the volume of the bubble
☐	**A**	decreases	decreases
☐	**B**	decreases	increases
☐	**C**	increases	decreases
☐	**D**	increases	increases

(ii) Which of these is a unit of pressure?

(1)

☐ **A** g/cm^3
☐ **B** J
☐ **C** kg/cm
☐ **D** Pa

(iii) The diver measures air pressure in atmospheres.

A bubble has an initial volume, V_1, of 0.50 litres, at a pressure, P_1, of 3.30 atmospheres.
The bubble rises towards the surface of the water, where the pressure, P_2, is 1.07 atmospheres.
Calculate the volume, V_2, of the bubble near the surface.
Use the equation

$$V_2 = \frac{P_1 \times V_1}{P_2}$$

(2)

volume, V_2, of the bubble = ... litres

(Total for question = 4 marks)

Q2.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The particles of a gas exert a pressure on the walls of a container.

Which row of the table is correct when the pressure of the gas changes?

(1)

	pressure of gas	number of particles colliding with the walls of the container each second
☐ **A**	increases	stays the same
☐ **B**	increases	increases
☐ **C**	decreases	stays the same
☐ **D**	decreases	increases

(Total for question = 1 mark)

Q3.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Figure 16 shows a metal container with a movable piston.

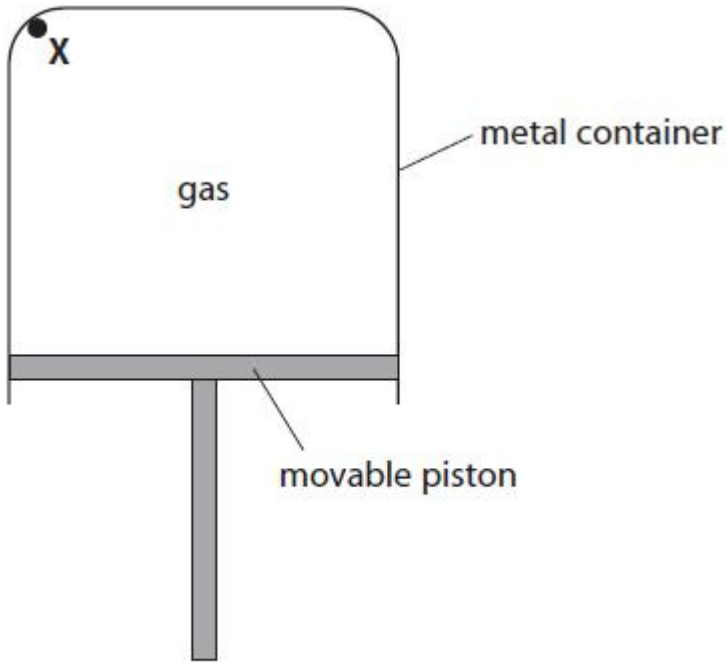


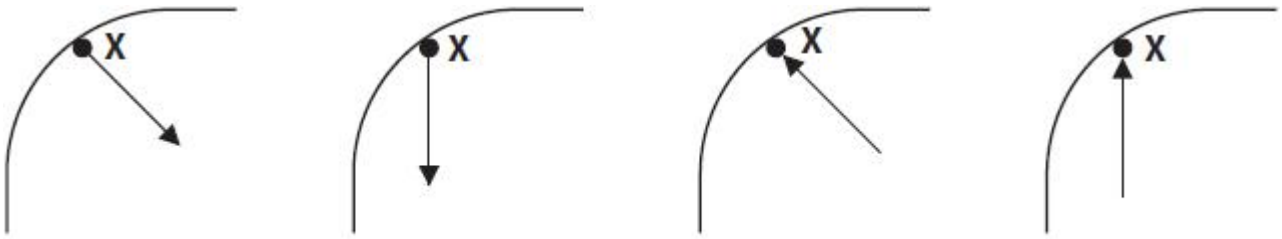
Figure 16

Point X is on the inner surface of the container.

The gas in the container is at a higher pressure than the air outside the container.

Which of these shows the direction of the force, due to the gas, on the container at point X?

(1)



☐ **A** ☐ **B** ☐ **C** ☐ **D**

(Total for question = 1 mark)

Q4.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of these means changing state from solid directly to gas?

(1)

☐ **A** condensing
☐ **B** freezing
☐ **C** melting
☐ **D** sublimating

(Total for question = 1 mark)

Q5.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Solid, liquid and gas are states of matter.

Which process describes the change from a solid to a liquid?

(1)

- ☐ **A** melting
- ☐ **B** freezing
- ☐ **C** evaporation
- ☐ **D** condensation

(Total for question = 1 mark)

Q6.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Which of these describes the motion of particles in a liquid?

(1)

- ☐ **A** particles move around and are very close together
- ☐ **B** particles move very quickly and are a long way apart
- ☐ **C** particles do not move and are a long way apart
- ☐ **D** particles vibrate about fixed positions

(Total for question = 1 mark)

Q7.

A student changes the volume of gas in a container and notes the pressure for different values of the volume.
The results are shown in Figure 6 and plotted on the graph in Figure 7.

volume in ml	pressure in kPa
10	260
12	200
20	140
25	150
30	100
40	75
50	65

Figure 6

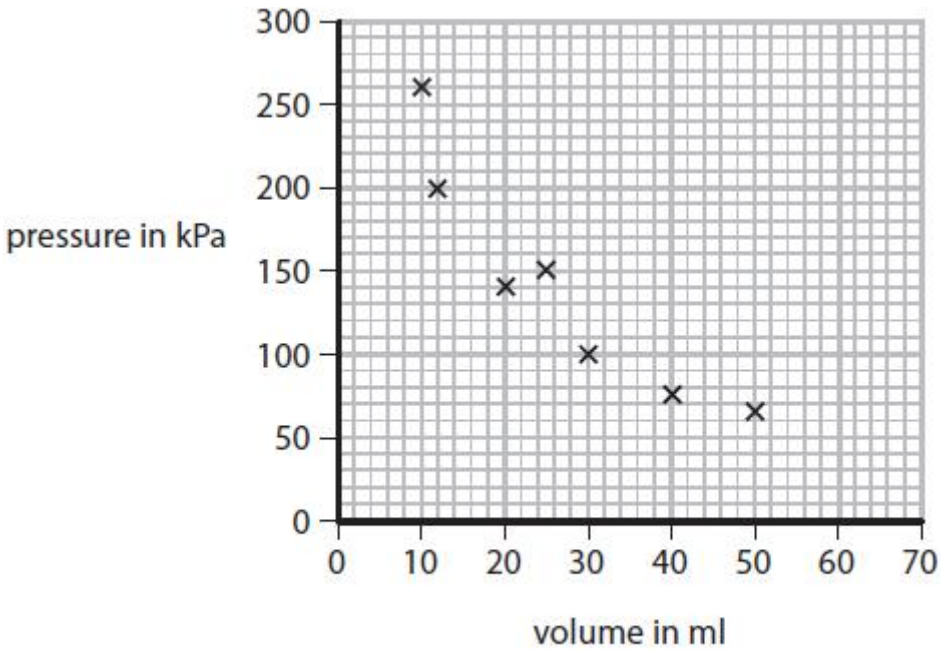


Figure 7

(i) Identify the anomalous result plotted on Figure 7 by drawing a circle on Figure 7 around the anomalous point.

(1)

(ii) Draw the curve of best fit on Figure 7.

(1)

(iii) Describe how the graph in Figure 7 would change if the student repeated the experiment with the same mass of gas, at a higher constant temperature.

(2)

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(Total for question = 4 marks)

Q8.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

A technician is investigating the pressure and volume of some gas trapped in a container.

The table in Figure 16 shows the results from the investigation.

pressure in kPa	volume in cm^3
100	270
110	245
130	208
150	180
170	159
190	142
210	129

Figure 16

Figure 17 is a graph of the results.

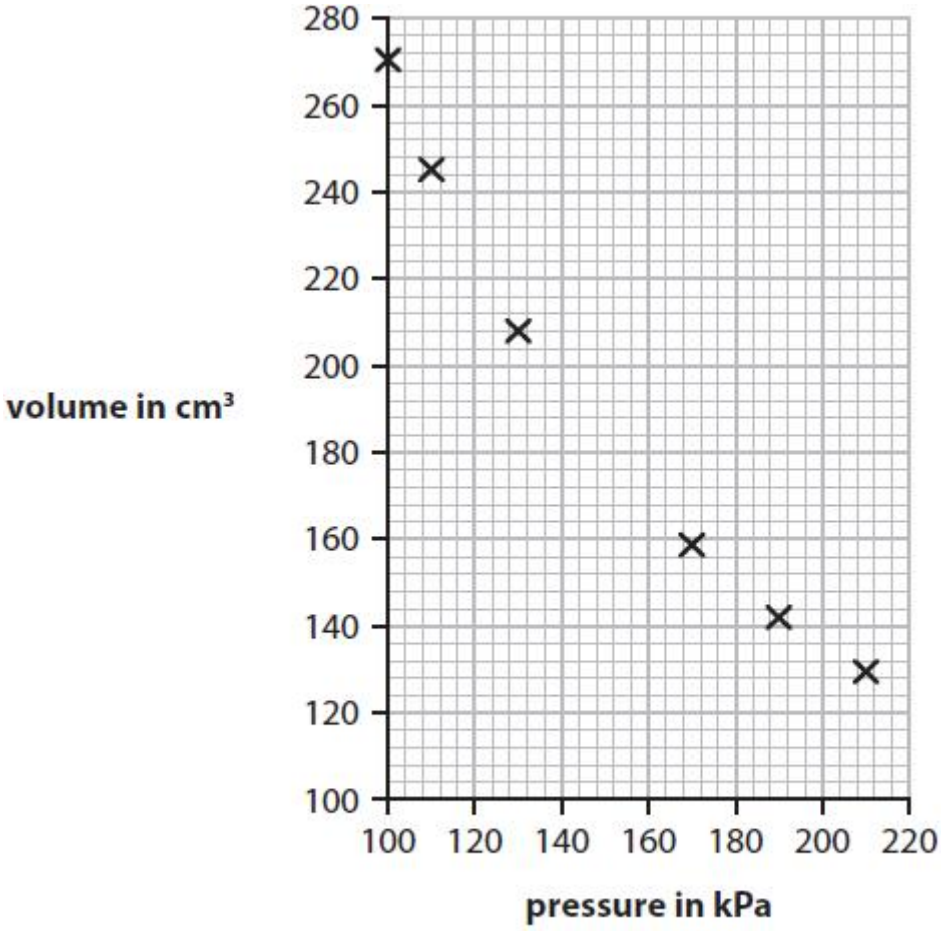


Figure 17

(i) One point has not been plotted on the graph in Figure 17.

The values for this point are shaded in the results table in Figure 16.
Plot the missing point on the graph in Figure 17.

(1)

(ii) Draw a smooth curve through the points on the graph in Figure 17.

(1)

(iii) Use the graph in Figure 17 to estimate the volume at a pressure of 120 kPa.

(1)

volume = ...cm^3

(iv) The temperature of the gas in the container is 293 K.

Which of these is the same temperature as 293 K?

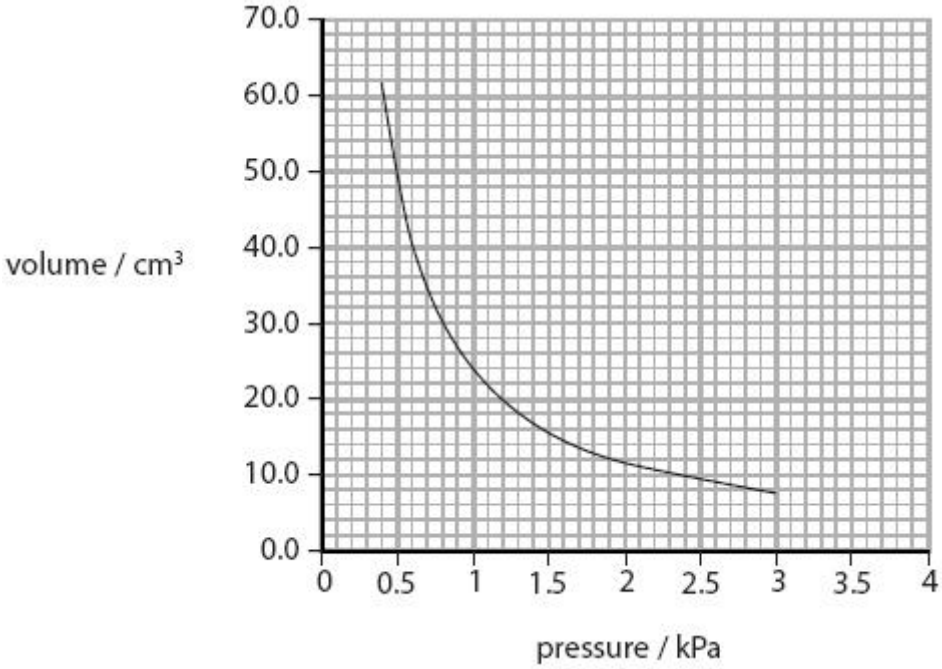
(1)

- ☐ **A** −20 °C
- ☐ **B** 0 °C
- ☐ **C** 20 °C
- ☐ **D** 273 °C

(Total for question = 4 marks)

Q9.

(a) A student investigated how the volume and pressure of a gas were related.
The graph shows how the volume of a gas changes with pressure.



The table shows the results used to plot the graph.

pressure / kPa	volume / cm³
2.5	.....
2.0	11.9
.....	14.0
1.4	17.0
1.0	24.0
0.4	61.5

(i) Use the graph to complete the table.

(2)

(ii) The results were taken at a constant temperature of 23 °C.
Complete the sentence by putting a cross (☒) in the box next to your answer.
A temperature of 23 °C can be written in kelvin as

(1)

☐ **A** −273 K

☐ **B** 250 K

☐ **C** 273 K

☐ **D** 296 K

(iii) Estimate a value for the volume when the pressure becomes 4 kPa.

(1)

volume =.....cm^3

(iv) When the pressure of the gas is 2.2 kPa, the volume of the gas is 10.8 cm^3.

Use the equation

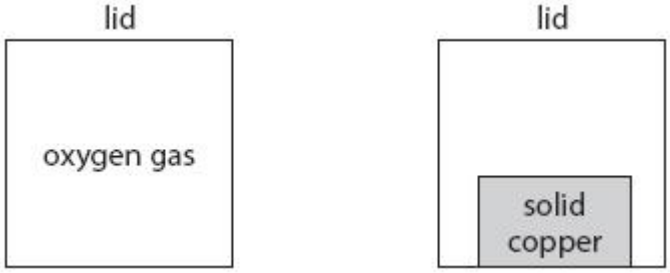
$$V_2 = \frac{P_1V_1}{P_2}$$

to calculate the volume of the gas when the pressure of the gas is 0.2 kPa.

(2)

volume =.....cm^3

*(b) The diagrams show a block of solid copper and some oxygen gas in two closed containers.



The oxygen exerts a pressure on the lid of its container.
The copper does not exert a pressure on the lid of its container.
Explain, using kinetic theory, why the oxygen exerts a pressure on the lid but the copper does not.

(6)

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(Total for Question = 12 marks)

Q10.

Figure 1 shows a fixed mass of gas inside a cylinder with a movable piston.

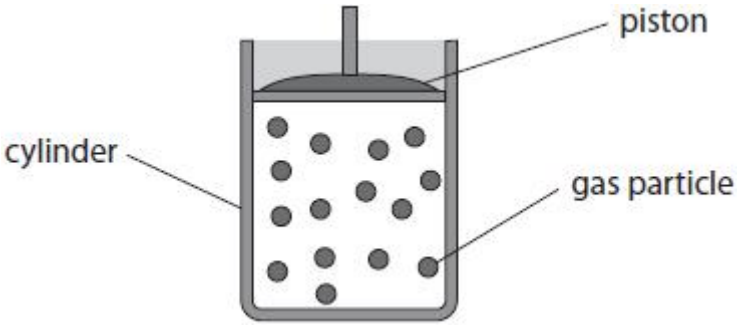

Figure 1

(i) Describe, in terms of **gas particles**, how the gas exerts a pressure on the cylinder.

(3)

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(ii) Figure 2 shows the same gas squashed into a smaller volume.

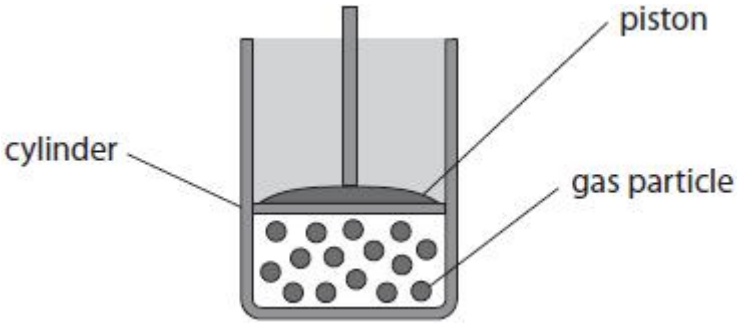

Figure 2

State what happens to the pressure the gas exerts on the cylinder when the volume of gas is reduced, as in Figure 2.

(1)

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.....

(iii) State what happens to the gas particles when the volume of the gas is reduced, as in Figure 2.

(1)

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.....

(Total for question = 5 marks)

Mark Scheme

Q1.

Question number	Answer	Mark
i	B decreases increases A is wrong because the volume of the bubble does not decrease C and D are wrong because the air pressure in the bubble does not increase	(1) AO1

Question number	Answer	Additional guidance	Mark
ii	**D** Pa A , B and C are wrong because they are all wrong units for pressure		(1) AO1

Question number	Answer	Additional guidance	Mark
iii	substitution (1) (V_2 =) $\frac{0.5 \times 3.3}{1.07}$ evaluation (1) (volume of the bubble =) 1.5	 any number that rounds to 1.5 (m/s) accept 1.54 award full marks for the correct answer without working	(2) AO2

Q2.

Question Number	Answer	Mark
(i)	**B** increase | increase **B** is the only correct answer. **A** is incorrect because as the pressure of the gas increases the number of particles colliding with the walls of the container does not stay the same. **C** is incorrect because as the pressure of the gas decreases the number of particles colliding with the walls of the container does not stay the same. **D** is incorrect because as the pressure of the gas decreases the number of particles colliding with the walls of the container does not increase.	(1)

Q3.

Question number	Answer	Additional guidance	Mark
	 B and D are incorrect because they are not normal to the surface A is incorrect because the force should act outwards		(1)

Q4.

Question number	Answer	Mark
	☒ D sublimating A is incorrect because it describes a change of state from gas to liquid. B is incorrect because it describes a change of state from liquid to solid C is incorrect because it describes a change of state from solid to liquid	**(1)** **AO1**

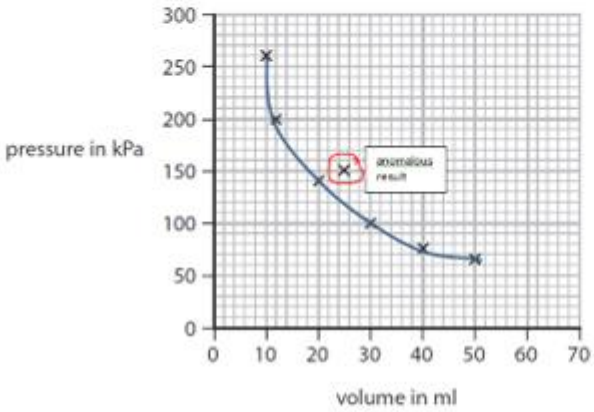
Q5.

Question Number	Answer	Mark
	A melting **A** is the only correct answer. **B** is incorrect because the change from solid to liquid is not freezing. **C** is incorrect because the change from solid to liquid is not evaporation. **D** is incorrect because the change from solid to liquid is not condensation.	**(1)**

Q6.

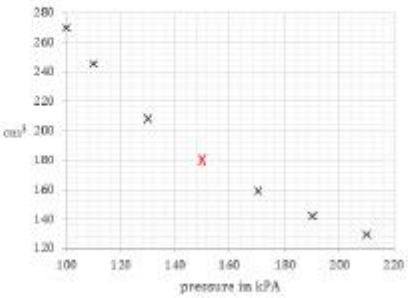
Question Number	Answer	Mark
	A particles move around and are very close together **B** is incorrect because describes particles in a gas **C** is incorrect because does not describe solid, liquid or gas as the particles always move **D** is incorrect because describe particles in a solid	**(1)** **AO1.1**

Q7.

Question Number	Answer	Additional guidance	Mark
			**(2)**
(i)	anomalous point (1)	ringed or other indication	
(ii)	curve touches one part of the cross for each of the points, excluding the anomalous point (1)	ignore curve beyond 260 kPa and beyond 50ml	

Question Number	Answer	Additional guidance	Mark
(iii)	A description that combines the following points the line will be higher (1) have a similar shape (1)	 Allow for one mark all data will be higher allow the pressure will be higher for the same volume for 2 marks allow the volumes will be higher for the same pressure for 2 marks	(2)

Q8.

Question	Answer	Additional guidance	Mark
(i)	point plotted at 150,180 (1)	± half a square 	(1) AO3.1

Question	Answer	Additional guidance	Mark
(ii)	smooth curve through the points (1)	smooth curve through/touching at least 5 crosses ignore slight shakiness in drawing do not accept tramlining (multiple curves)	(1) AO3.1

Question	Answer	Additional guidance	Mark
(iii)	224 ± 4 (cm^3) (1)	any number between 220 and 228 inclusive	(1) AO3.1

Question	Answer	Mark
(iv)	**C** 20 °C ***A*** *is incorrect as it is 253 K* ***B*** *is incorrect as it is 273 K* ***D*** *is incorrect as it is 546 K*	(1) AO1.1

Q9.

	Answer	Acceptable answers	Mark
(a)(i)	volume in range 9.0 – 10.5 (cm^3) (1) pressure in range 1.5 – 1.7 (kPa) (1)		(2)
(a)(ii)	**D** 296 K		(1)
(a)(iii)	Volume in range 4 – 8 (cm^3)	Any value between 4 (cm^3) and 8 (cm^3)	(1)
(a)(iv)	Substitution (1) 2.2 × 10.8 ÷ 0.2 Evaluation (1) 119 (cm^3)	118.8 (cm^3) give full marks for the correct answer, no working	(2)

		Indicative Content	Mark
QWC	*(b)	An explanation including some of	(6)

		the following points: particles in gas • move rapidly • throughout container • collide with each other • collide with walls/lid of container • exerting a forceparticles in solid • in fixed positions • vibrate • do not reach lid	
Level	**0**	No rewardable content	
1	**1 - 2**	• a limited explanation e.g. particles in the copper do not touch the lid / particles in the oxygen do touch the lid. • the answer communicates ideas using simple language and uses limited scientific terminology • spelling, punctuation and grammar are used with limited accuracy	
2	**3 - 4**	• a simple explanation e.g. particles in a gas can move freely and collide with the lid • the answer communicates ideas showing some evidence of clarity and organisation and uses scientific terminology appropriately • spelling, punctuation and grammar are used with some accuracy	
3	**5 - 6**	• a detailed explanation e.g. particles in a gas can move freely and collide with the lid but particles in a solid vibrate about fixed positions so cannot reach the lid • the answer communicates ideas clearly and coherently uses a range of scientific terminology accurately • spelling, punctuation and grammar are used with few errors	

Q10.

Question Number:	Answer	Additional guidance	Mark
(i)	a description to include 3 of the following: gas particles/molecules/atoms are continually in motion (1) collide /hit/bombard (1) with the (sides/walls) of the container (1) (gas particles) exert force (on the side of the container) (1)	allow cylinder /glass	(3) AO 1 1

Question Number:	Answer	Additional guidance	Mark
(ii)	pressure (of gas) increases (1)		(1) AO 1 1

Question Number:	Answer	Additional guidance	Mark
(iii)	more (frequent) collisions (of particles with walls) (1)	accept particles move closer together particles move faster	(1) AO 1 1